Hope Enterprise University College	
Faculty of Informatics	
Department of Computer Science	
Course Plan	
Course Title	**Introduction to Artificial Intelligence**
Course Code:	**CS 443**
Credit Hour	**3**
Contact Hour	**4**
Lab Hours	**2**
Prerequisite	**Data structure and Algorithm Design**
Course Category	**Major**
Year	**4**
Semester	**1**
Instructor	**Ashenafi Worku**

Course Description

This course introduces to the students' basic principles and concepts in the area of Artificial Intelligence. It covers topics like different approaches to AI, historical foundation of AI, major application areas for AI, different agent structures applicable in different environments, methods used to formally represent real world problems, search of problem spaces for solutions, and deduction of knowledge in terms of logic and reasoning. Application of these methods is made to important areas of Artificial Intelligence including Expert Systems, Natural language processing, machine learning, neural networks, computer vision and robotics. The course enable student to get general idea about these different areas of AI application and challenges in AI field.

Course Objectives

Upon completion of this course students should be able to:

- Understand meaning and historical development of AI
- Describe the key components of the artificial intelligence (AI) field
- Know different agent environments.
- Outline search strategies and fine problem's applying a suitable search method
- Grasp how agents reasoning works
- Know the different agent structures and t.

- Understand knowledge representation methods used in AI
- Use prolog to build knowledge base.

Course Content	Weeks
Chapter 1: Introduction to Artificial Intelligence	
1.1 Definition and approaches in AI 1.2 Foundations of Artificial Intelligence 1.3 State of the art in AI 1.4 History of AI	**1- 2**
Chapter 2: Intelligent Agents (8 hours/2weeks)	
2.1 Agent and Agent Environments 2.2 Types of Agent programs 2.2.1 Simple reflex agent 2.2.2 Model based reflex agent 2.2.3 Goal based agent 2.2.4 Utility based agent 2.2.5 Learning Agent	**3 - 4**
Chapter 3: Problem Solving by Searching – Uninformed Search Strategies	**5 - 8**
3.1 Problem formulation 3.2 Searching for solution 3.3 Search Strategies 3.3.1 Uninformed Search strategies 3.3.1.1 Breadth first 3.3.1.2 Uniform cost 3.3.1.3 Depth first 3.3.1.4 Depth limited 3.3.1.5 Iterative deepening 3.3.2 Comparison among Uninformed Search Strategies	
Mid Test	**9**
Chapter 4: Problem Solving by Searching – Informed Search Strategies	**10 - 13**

4.1 Informed Search strategies 4.2 Best first search, 4.2.1.1 Greedy Best First Search 4.2.1.2 A* search 4.2.1.3 Memory Bounded A* Search 4.2.2 Comparison among Informed Search Strategies 4.3 Other Search strategies	
Chapter 5 : Logical Agents - Knowledge Based System	**14- 16**
5.1 Knowledge based agents 5.2 Logic 5.2.1 Prepositional Logic 5.2.1.1 Propositional Logic Theorem Proving 5.2.1.2 Propositional Model Checking 5.2.1.3 Agent based on propositional logic 5.2.2 First Order Logic 5.2.2.1 Syntax and semantics of FOL 5.2.2.2 Knowledge Engineering in FOL 5.2.2.3 Inference in FOL	
Lab Session – Introduction to Python	5 - 16

Module 1: Introduction -- Relationship between computers and programs -- Basic principles of computers -- File systems -- Using the Python interpreter -- Introduction to binary computation -- Input / Output **Module 2: Data types and control structures** -- Operators (unary, arithmetic, etc.) -- Data types, variables, expressions, and statements -- Assignment statements -- Strings and string operations -- Control Structures: loops and decision **Module 3: Modularization and Classes** -- Standard modules – Packages -- Defining Classes -- Defining functions -- Functions and arguments (signature) **Module 4: Exceptions and data structures** -- Data Structures (array, List, Dictionary) -- Error processing -- Exception Raising and Handling	(2hrs per week)

Teaching and Learning Method:

Lecture, Group Discussion, Laboratory practical exercise, Individual and group assignments and Class exercise

Assessment /Evaluation Scheme

Component	**Weight**	**Coverage**	**Assessment Time**
Individual assignment-1	5%	Chapter 2	Week 3
Individual assignment-2	5%	Chapter 3 &4	Week 5
Mid test	**20%**	**Chapter 1 & 2,3**	**Week 8**
Group Assignment 1	5%	Lab Lesson	Week 11

Lab practical test -1	5%	Unformed and Uniformed search strategy	Week 13
Lab practical test -2	10%	All lab class topics	Week 15
Final Exam	**50%**	**All chapters**	
Textbook: Russell, S and Peter Norvig. Artificial Intelligence: A Modern Approach (3rd edition). Cambridge University Press, 2004			
Reference books: 1. Yao, Mariya, Adelyn Zhou and Marlene Jia . Applied Artificial Intelligence: A Handbook For Business Leaders Paperback . ,2018 2. Ertel, Wolfgang and Nathanael T. Black . Introduction to Artificial Intelligence, 2nd ed. 2017 3. Object_oriented_python_tutorial 4. Hans-Petter Halvorsen ,Python Programming			